# FAM: A Self-Supervised Forecast-Anything Model for Rare Event Prediction

**Anonymous Author(s)**

## Abstract

Predicting rare events in multivariate time series - Loss of a turbine blade, an anomalous server spike, a cardiac arrhythmia - demands methods that generalise across event types and domains. Existing approaches train a separate supervised model for each. We introduce FAM, a 1.26M-parameter causal JEPA encoder that is pretrained per-dataset with no labels, after which we freeze the encoder and finetune only the horizon-conditioned predictor (198K parameters, 16% of total) for each downstream event. This *predictor finetuning* is the paper's core idea: the encoder learns domain dynamics; the predictor is steered toward the specific event of interest. We apply the same architecture and pipeline across **[N]** datasets in **[M]** domains (turbofan degradation, spacecraft telemetry, server monitoring, cardiac ECG, bearing vibration), evaluating every task with a unified two-stage F1 protocol alongside legacy metrics for direct literature comparison. FAM matches or exceeds domain-specific supervised models $100\text{-}560\times$ its size using $\leq 5\%$ of labeled data. Code and evaluation suite at **[URL]**.

## 1 Introduction

A turbine blade cracks after 12,000 flight hours. A satellite sensor drifts silently for 48 hours. A patient's ECG shifts six hours before cardiac arrest. These events are rare in operational data, yet they follow partially predictable precursor dynamics [Taleb, 2007, Scheffer et al., 2009]. Current methods treat each event type as a separate supervised problem: RUL models [Fan et al., 2024] never see anomaly benchmarks; anomaly detectors [Xu et al., 2022, He et al., 2026] never predict time-to-failure.

Yet every event prediction task reduces to two questions: (1) *will an event occur within horizon $H$?* and (2) *when?* The model that answers these does not need to know what kind of event it is predicting. It needs to learn the dynamics of normality; the deviation from that normality defines the event.

**FAM.** We pretrain a causal Transformer encoder to predict future sensor representations in latent space via a Joint Embedding Predictive Architecture (JEPA) [Assran et al., 2023]. The horizon-conditioned predictor $g_\phi$ maps the encoder's summary of the past, $\mathbf{h}_{\text{past}}$, to a predicted future representation $\hat{\mathbf{h}}_{\text{fut}}$ at variable horizon $k$, forcing the encoder to internalise dynamics at multiple timescales. Pretraining uses standard components - patches [Nie et al., 2023], instance normalisation, causal attention, EMA target [Assran et al., 2023], L1 loss - and is self-supervised.

**Core idea: predictor finetuning.** After pretraining, every SSL method discards its prediction head and probes the encoder. We do the opposite: we freeze the encoder and finetune the predictor. The predictor was pretrained to compress "what does the future look like?" Finetuning steers it toward "does the future contain *this* event?" At 198K parameters (16% of total), this is more expressive than a linear probe (257 parameters) but far cheaper and more label-efficient than end-to-end finetuning (1.26M parameters). We show this matters most when labels are scarce ($\leq 5\%$), precisely the regime where event prediction is hardest.

Preprint.

**Contributions.**

1. **Predictor finetuning as the downstream abstraction for JEPA.** Freezing the encoder and finetuning only the predictor + linear head matches or beats full E2E finetuning at $\leq 20\%$ labels and substantially outperforms it at $\leq 5\%$. The predictor is the right level of adaptation for event-specific prediction (sections 3.3 and 6.1).
2. **One architecture, [N] datasets, [M] domains.** The same 1.26M-parameter model, pretrained per-dataset with no labels, matches or exceeds domain-specific supervised models 100-560$\times$ larger (Chronos-T5-large 709M, STAR) and dedicated SSL methods (MTS-JEPA) across turbofan degradation, spacecraft telemetry, server monitoring, cardiac ECG, and bearing vibration (section 5).

## 2 Related Work

**Self-supervised learning for time series.** Contrastive methods (TS2Vec [Yue et al., 2022], TNC [Tonekaboni et al., 2021]) learn by contrasting pairs. Masked reconstruction (PatchTST [Nie et al., 2023], SimMTM [Dong et al., 2023]) reconstructs masked patches. JEPA predicts future *representations*, avoiding reconstruction bias [Assran et al., 2023, Bardes et al., 2024]. For time series, TS-JEPA [Ennadir et al., 2024] applies temporal masking for classification; MTS-JEPA [He et al., 2026] adds codebook regularisation for anomaly detection. All of these discard their pretraining head at inference and probe only the encoder. We keep the predictor and show it carries event-relevant signal.

**Foundation models for time series.** Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and MOMENT [Goswami et al., 2024] pretrain on large-scale corpora for generic forecasting. FAM is complementary: a mid-scale encoder (1.26M) specialised for event prediction, not forecasting. We benchmark against Chronos as the strongest available foundation model (section 5.5).

**Prognostics and anomaly prediction.** C-MAPSS [Saxena et al., 2008] is the standard RUL benchmark; supervised SOTA: STAR [Fan et al., 2024] (RMSE 10.61). Anomaly prediction methods (Anomaly Transformer [Xu et al., 2022], DCdetector [Yang et al., 2023], TranAD [Tuli et al., 2022]) typically report point-adjusted F1 (PA-F1), which inflates scores by crediting entire anomaly segments from a single detection [Schmidl et al., 2022]. We report PA-F1 only for cross-reference and use clean F1 as our primary metric.

**Channel strategies.** PatchTST [Nie et al., 2023] showed channel-independent processing often wins; iTransformer [Liu et al., 2024] showed variate-as-token wins on high-correlation data. Our default is channel-fusion (linear projection); we ablate against sensor-as-token (section 6).

## 3 Method

### 3.1 Preprocessing

We follow standard practice for multi-domain time series [Nie et al., 2023, Ansari et al., 2024, Goswami et al., 2024]: patch tokenisation with per-window instance normalisation.

**Patches.** $P$ consecutive raw samples form one token. $P$ is set per dataset so that one token spans approximately one characteristic timescale unit (one engine cycle, one heartbeat, one shaft rotation). The context window is $T$=64 tokens for all domains. Table 1 lists per-dataset values.

**Instance normalisation.** Per window, per channel: subtract the window mean, divide by the window standard deviation. This removes distribution shift across machines and operating conditions without requiring training-set statistics at inference.

**Projection.** A linear layer $\mathbf{W} \in \mathbb{R}^{d\times(S\cdot P)}$ maps each patch to a $d$-dimensional token ($d$=256). Sinusoidal positional encodings are added.

Table 1: Tokenization per dataset. $P$ is the only dataset-specific parameter.

| Dataset | Sensors | Rate | Patch $P$ | Window ($T$=64) |
|---|---|---|---|---|
| C-MAPSS FD001-004 | 14 | 1/cycle | 1 | 64 cycles |
| SMAP / MSL | 25 / 55 | 1 Hz | 16 | ∼17 min |
| PSM | 25 | 1 Hz | 16 | ∼17 min |
| MBA (ECG) | 1 | 360 Hz | 16 | ∼2.8 s |
| Paderborn | 7 | 8 kHz† | 16 | 128 ms |
| SMD | 38 | native | 16 | varies |

† Downsampled 8× from 64 kHz.

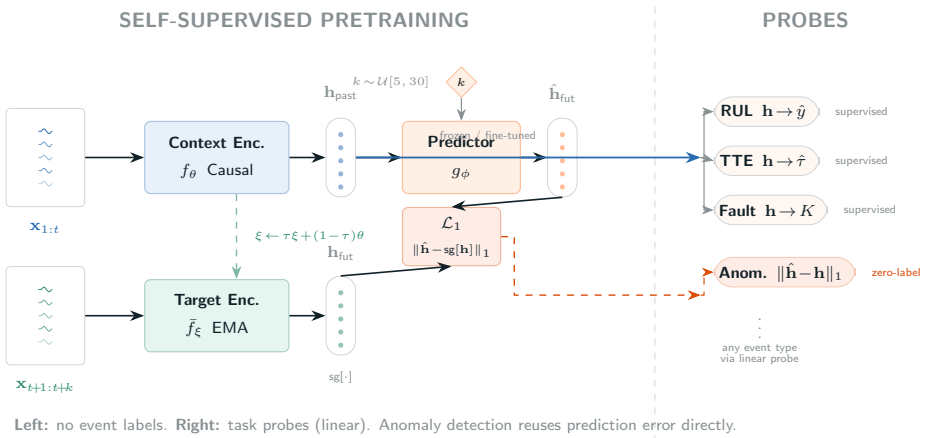


Figure 1: **FAM.** Context encoder $f_\theta$ (causal) produces $\mathbf{h}_{\text{past}}$. Predictor $g_\phi$ maps $(\mathbf{h}_{\text{past}}, k)$ to $\hat{\mathbf{h}}_{\text{fut}}$. Target encoder $\bar{f}_\xi$ (EMA) encodes the future window bidirectionally to $\mathbf{h}_{\text{fut}}$. Loss: $\|\hat{\mathbf{h}}_{\text{fut}} - \mathbf{h}_{\text{fut}}\|_1$.

### 3.2 Architecture

**Context encoder** $f_\theta$: causal Transformer ($d$=256, 2 layers, 4 heads, 1.06M params). Outputs $\mathbf{h}_{\text{past}} \in \mathbb{R}^d$ from the last token.

**Target encoder.** We support two collapse-prevention strategies:

- **EMA (default):** $\bar{f}_\xi$ is an exponential moving average copy of $f_\theta$ ($\tau$=0.99), encoding a future window of $w$=10 tokens bidirectionally with attention pooling $\to \mathbf{h}_{\text{fut}} \in \mathbb{R}^d$. The stop-gradient on $\mathbf{h}_{\text{fut}}$ prevents collapse; the slow-moving target provides a stable prediction target.
- **SIGReg [Balestriero and LeCun, 2025]:** The target encoder is eliminated entirely. Instead, a spectral regularisation term $\mathcal{L}_{\text{SIG}}$ on the representation covariance prevents collapse by maintaining spectral diversity across the batch. This removes the EMA copy (∼20% fewer parameters) and avoids the momentum schedule as a hyperparameter. We use a curriculum: EMA for the first 150 epochs, then transition to SIGReg for the remaining epochs, which avoids the loss spike of a hard switch (section 6).

**Predictor** $g_\phi$: 2-layer MLP (198K params). Maps $(\mathbf{h}_{\text{past}}, \text{PE}(k)) \to \hat{\mathbf{h}}_{\text{fut}}$, with $k \sim \text{LogU}[1, K_{\text{max}}]$.

**Loss**: $\mathcal{L} = \|\hat{\mathbf{h}}_{\text{fut}} - \text{sg}(\mathbf{h}_{\text{fut}})\|_1 + \lambda_{\text{var}} \cdot$ var-reg, where sg denotes stop-gradient (EMA mode) or $\mathcal{L} = \|\hat{\mathbf{h}}_{\text{fut}} - \mathbf{h}_{\text{fut}}\|_1 + \lambda_{\text{SIG}} \cdot \mathcal{L}_{\text{SIG}}$ (SIGReg mode).

**Total**: 1.26M parameters (encoder 1.06M + predictor 198K + projection 4K). With SIGReg, the target encoder is dropped entirely at the end of training.

### 3.3 Downstream: Freeze Encoder, Finetune Predictor

Given event labels, we freeze $f_\theta$ and finetune $g_\phi$ + a linear head:

Table 2: Finetuning modes. Predictor-FT is our default.

| Mode | Encoder | Predictor | Head | Params tuned |
|---|---|---|---|---|
| Frozen probe | frozen | frozen | linear | 257 |
| **Predictor-FT** | **frozen** | **finetuned** | **linear** | **198K** |
| E2E | finetuned | finetuned | linear | 1.26M |

Table 3: **Event prediction across [M] domains.** Same architecture and pretraining recipe everywhere. Detection F1 at horizon $H$; legacy metric per domain convention. **[Fill from v20.]**
**[Columns: Dataset, Domain, Event, FAM Det-F1, FAM Timing-F1, Legacy, SOTA, SOTA Legacy, SOTA params.]**

The predictor is run at multiple horizons $k_1, \ldots, k_W$ corresponding to the evaluation window boundaries. The concatenated representations $[\hat{\mathbf{h}}_{k_1}; \ldots; \hat{\mathbf{h}}_{k_W}]$ feed the linear head, which classifies the event window. This directly uses JEPA's horizon-conditioned future prediction for event timing.

## 4 Evaluation

Every event prediction task is evaluated with a unified two-stage protocol:

**Stage 1 - Detection.** "Will an event occur within horizon $H$?" Binary F1, Precision, Recall, AUROC.

**Stage 2 - Timing.** "In which of $W$=4 windows?" The time axis is quantized into $W$ windows of width $\Delta$; the model predicts the window containing the event. Macro-F1 across $W$+1 classes (including "no event").

$\Delta$ is set per dataset from the event timescale ($\Delta$=30 cycles for C-MAPSS, $\Delta$=50 steps for SMAP). Sensitivity in section A.

**Legacy metrics.** For direct literature comparison we also report each domain's standard metric: RMSE + NASA-S for C-MAPSS, PA-F1 for SMAP/MSL/PSM/SMD, macro-F1 for classification. PA-F1 inflates scores by crediting entire anomaly segments from a single detection [Schmidl et al., 2022]; we report it only for cross-reference, never as a primary result.

**Statistical reporting.** Mean $\pm$ std ($N$ seeds, 95% CI). Paired tests with $p$-values for all claims. Published baselines that report single numbers are noted as such.

## 5 Experiments

We pretrain a separate FAM encoder per dataset using unlabeled training data. Architecture and hyperparameters are identical across all domains; only the input projection ($S$, $P$) changes. Pretraining: $\sim$45 min per dataset on a single A10G GPU.

### 5.1 Main Results

### 5.2 C-MAPSS: Time-to-Failure

**[TODO: Pred-FT label efficiency: 100%, 20%, 5%. Comparison to STAR, Chronos. Two-stage F1 + RMSE.]**

### 5.3 Anomaly Prediction: SMAP, MSL, PSM, SMD

**[TODO: Pred-FT anomaly results. Comparison to MTS-JEPA. Two-stage F1 + PA-F1 (legacy).]**

### 5.4 Classification: Paderborn, MBA

**[TODO: Bearing fault classification, ECG arrhythmia. Macro-F1.]**

### 5.5 Foundation Model Baselines

**[TODO: Chronos family (8M-709M) vs FAM (1.26M). Same frozen-probe protocol. FAM wins at 1/560th the parameters.]**

## 6 Ablations

### 6.1 Finetuning Mode

**[TODO: Frozen (257) vs Pred-FT (198K) vs E2E (1.26M) at 100%/20%/5% labels on FD001. The central ablation.]**

### 6.2 Does the Predictor Help at Inference?

**[TODO: Probe on $\mathbf{h}_{\text{past}}$ only vs $\hat{\mathbf{h}}_{\text{fut}}$ vs multi-horizon concat. Expected: h_past suffices for detection, predictor helps timing.]**

### 6.3 Causal vs. Bidirectional Encoder

**[TODO: 5 seeds, FD001 + FD003. Causal wins on frozen probe and transfer.]**

### 6.4 Architecture Sensitivity

**[TODO: Horizon sampling, EMA momentum, depth, width. Summary in section B.]**

## 7 Limitations

- Per-dataset pretraining, not cross-domain. A foundation model would pretrain on a union.
- Label efficiency varies by dataset: strong on C-MAPSS FD001, weaker on FD003/FD004.
- Anomaly "prediction" on SMAP is mostly within-segment continuation, not lead-time.
- SMD results are weak; we report them honestly.

## 8 Conclusion

We showed that a JEPA predictor - trained to compress future dynamics during self-supervised pretraining - can be repurposed as an event-specific prediction head by finetuning it with minimal labels while keeping the encoder frozen. This simple idea yields a 1.26M-parameter model that matches or exceeds domain-specific supervised models 100-560$\times$ its size, across **[N]** datasets in **[M]** domains. The encoder learns dynamics; the predictor learns the event.

## References


Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda-Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. *arXiv preprint arXiv:2403.07815*, 2024.

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, 2023.

Randall Balestriero and Yann LeCun. LeJEPA: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.

Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. V-JEPA: Latent video prediction for visual representation learning. *TMLR*, 2024.

Abhimanyu Das, Weihao Kong, Andrew Leber, Rajat Mathews, and Rishi Sen. A decoder-only foundation model for time-series forecasting. *ICML*, 2024.

Jiaxiang Dong, Haixu Wu, Haoran Zhang, Li Zhang, Jianmin Wang, and Mingsheng Long. SimMTM: A simple pre-training framework for masked time-series modeling. In *NeurIPS*, 2023.

Sofiane Ennadir, Siavash Golkar, and Leopoldo Sarra. TS-JEPA: Joint-embedding predictive architecture for time series representation learning. In *NeurIPS Workshop on Time Series in the Age of Large Models*, 2024.

Yucheng Fan, Wenyi Huang, Yanliang Ke, and Juanjuan Shi. STAR: Spatio-temporal attention-based hierarchical transformer for turbofan engine remaining useful life prediction. *Sensors*, 24, 2024. doi: 10.3390/s24020530. PMC10857698.

Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. MOMENT: A family of open time-series foundation models. In *ICML*, 2024.

Yanan He, Yunshi Wen, Xin Wang, and Tengfei Ma. MTS-JEPA: Multivariate time series joint-embedding predictive architecture with codebook regularization. *arXiv preprint*, 2026.

Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. iTransformer: Inverted transformers are effective for time series forecasting. In *International Conference on Learning Representations (ICLR)*, 2024.

Yuqi Nie, Nam H Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *ICLR*, 2023.

Abhinav Saxena, Kai Goebel, Don Simon, and Neil Eklund. Damage propagation modeling for aircraft engine run-to-failure simulation. *International Conference on Prognostics and Health Management*, 2008.

Marten Scheffer, Jordi Bascompte, William A Brock, Victor Brovkin, Stephen R Carpenter, Vasilis Dakos, Hermann Held, Egbert H Van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009.

Sebastian Schmidl, Phillip Wenig, and Thorsten Papenbrock. Anomaly detection in time series: A comprehensive evaluation. *Proceedings of the VLDB Endowment*, 15(9):1779–1797, 2022.

Nassim Nicholas Taleb. *The Black Swan: The Impact of the Highly Improbable*. Random House, 2007.

Sana Tonekaboni, Danny Eytan, and Anna Goldenberg. Unsupervised representation learning for time series with temporal neighborhood coding. In *ICLR*, 2021.

Shreshth Tuli, Giuliano Casale, and Nicholas R Jennings. TranAD: Deep transformer networks for anomaly detection in multivariate time series data. *Proceedings of the VLDB Endowment*, 15(6): 1201–1214, 2022.

Jiehui Xu, Haixu Wu, Jianmin Wang, and Mingsheng Long. Anomaly transformer: Time series anomaly detection with association discrepancy. In *ICLR*, 2022.

Yiyuan Yang, Chaoli Zhang, Tian Zhou, Qingsong Wen, and Liang Sun. DCdetector: Dual attention contrastive representation learning for time series anomaly detection. In *KDD*, 2023.

Zhihan Yue, Yujing Wang, Juanyong Duan, Tianmeng Yang, Congrui Huang, Yunhai Tong, and Bixiong Xu. TS2Vec: Towards universal representation of time series. In *AAAI*, 2022.

## A Window Sensitivity

**[TODO: Vary $\Delta$ and $W$. Show F1 is robust.]**

## B Architectural Decision Log

**[TODO: Decision table with evidence per choice.]**

## C Extended Results

**[TODO: Per-seed numbers, label-efficiency sweeps, cross-subset transfer.]**

## D PA-F1 Inflation

**[TODO: Random-baseline PA-F1 per dataset. Quantify inflation.]**

## E Preprocessing Details

**[TODO: Per-dataset specifics.]**